

Lab Report 1: Install Gitea with Docker and Test LFS Support

202383910018 TangWenting 唐雯婷

1. Objective

To learn how to install and use Docker Desktop

To deploy Gitea using Docker on a local machine

To understand and use Git Large File Storage (Git LFS)

To verify whether Gitea supports LFS

To create and manage a repository containing large files

2. Experimental Environment

Operating System: Windows

Docker: Docker Desktop

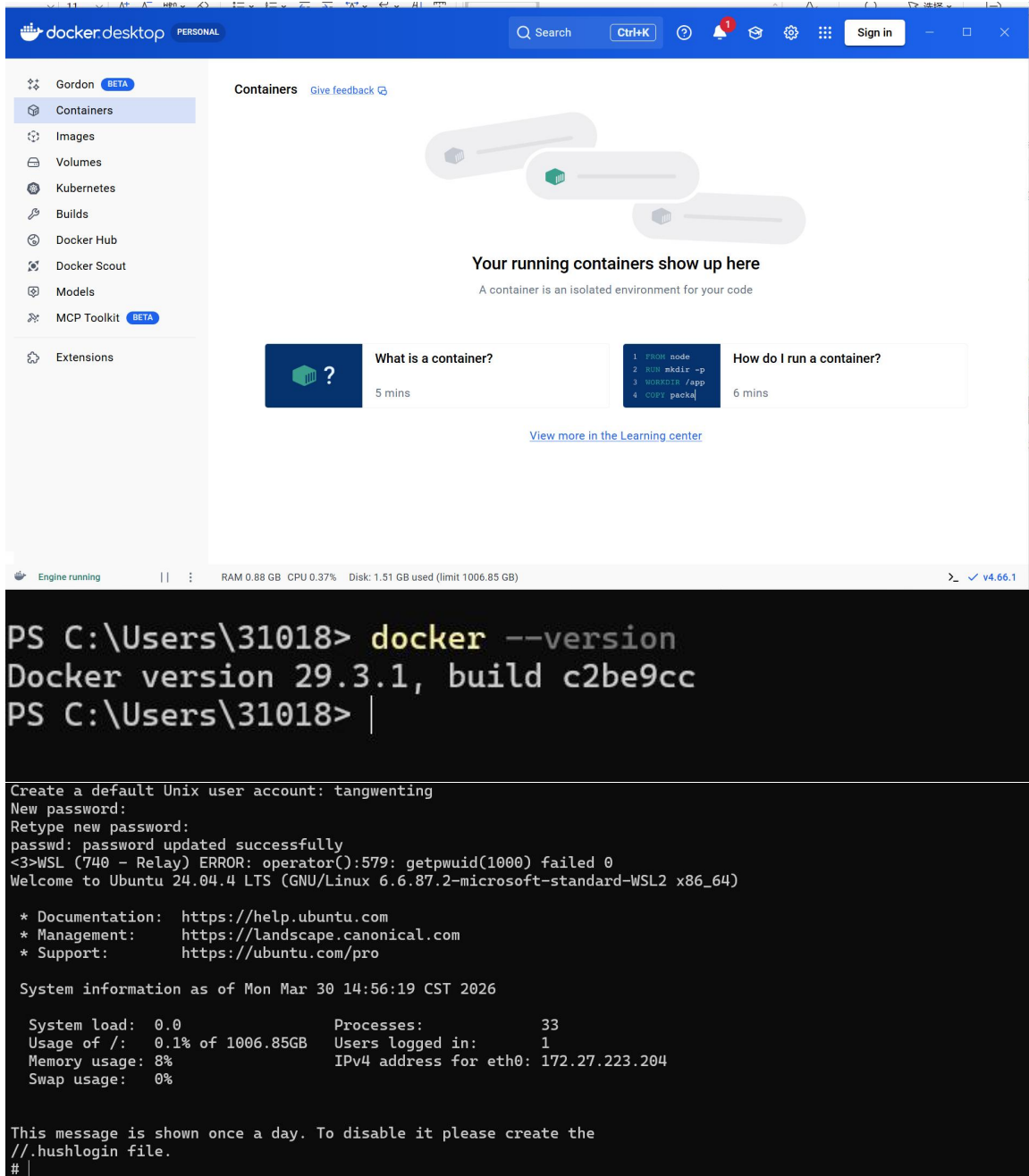
Version Control Tool: Git

Large File Tool: Git LFS

Platform: Gitea (running locally via Docker)

3. Experimental Content

Step 1: Install Docker Desktop (run 'wsl --install', install Docker Desktop, login)



Step 2: Deploy Gitea using Docker (docker run command, access http://localhost:3000)

```
PS C:\Users\31018> docker run -d --name gitea -p 3000:3000 -p 222:22 gitea/gitea:latest
Unable to find image 'gitea/gitea:latest' locally
latest: Pulling from gitea/gitea
0db3a2dc39b9: Pull complete
d49a2dee86fb: Pull complete
63ceb4d62a0b: Pull complete
cc4398cab299: Pull complete
5db3f65dd9c5: Pull complete
a0be84997a4f: Pull complete
edafa89f3509: Download complete
Digest: sha256:f846d26a4fc389c5806a580a765e00bfdd1fd181e6f2060da98ea2669d914472
Status: Downloaded newer image for gitea/gitea:latest
77ffb6e813b4d902bd5774a8e1ec44b1b911c3d01405d4142c7e45918dd24229
PS C:\Users\31018> |

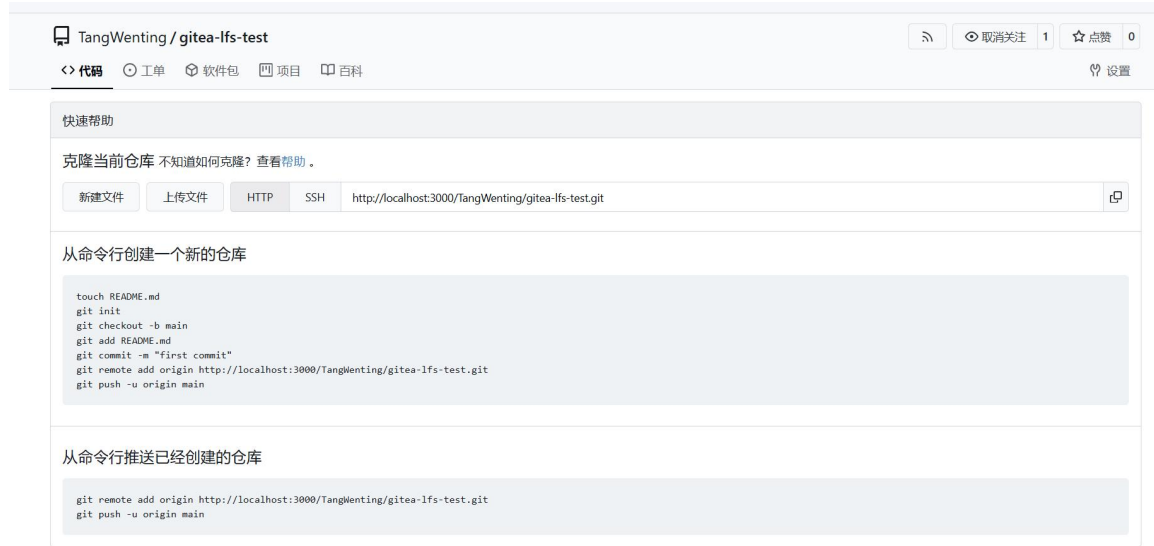
PS C:\Users\31018> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
77ffb6e813b4   gitea/gitea:latest  "/usr/bin/entrypoint..." About a minute ago Up About a minute  0.0.0.0:3000->3000
PS C:\Users\31018> |
```



Step 3: Initialize Gitea (complete setup and login)



Step 4: Create a repository (e.g., gitea-lfs-test)



Step 5: Clone repository (git clone and cd)

```
PS C:\Users\31018> git clone http://localhost:3000/tangwenting/gitea-lfs-test.git
Cloning into 'gitea-lfs-test'...
warning: You appear to have cloned an empty repository.
PS C:\Users\31018> cd gitea-lfs-test
PS C:\Users\31018\gitea-lfs-test> git clone http://localhost:3000/tangwenting/gitea-lfs-test.git
```

Step 6: Install and configure Git LFS (git lfs install, git lfs track '*.bin')

```
PS C:\Users\31018\gitea-lfs-test> git lfs install
Updated Git hooks.
Git LFS initialized.
PS C:\Users\31018\gitea-lfs-test> git lfs track "*.bin"
Tracking "*.bin"
PS C:\Users\31018\gitea-lfs-test> git add .gitattributes
PS C:\Users\31018\gitea-lfs-test> git commit -m "enable lfs"
[main (root-commit) 085d4a9] enable lfs
1 file changed, 1 insertion(+)
create mode 100644 .gitattributes
PS C:\Users\31018\gitea-lfs-test> |
```

Step 7: Create a large file (fsutil file createnew bigfile.bin 1073741824)

```
PS C:\Users\31018\gitea-lfs-test> fsutil file createnew bigfile.bin 1073741824
已创建文件 C:\Users\31018\gitea-lfs-test\bigfile.bin
PS C:\Users\31018\gitea-lfs-test> |
```

Step 8: Upload file (git add, commit, push)

```

PS C:\Users\31018\gitea-lfs-test> git add .
error: 'gitea-lfs-test/' does not have a commit checked out
fatal: adding files failed
PS C:\Users\31018\gitea-lfs-test> git commit -m "add big file"
On branch main
Your branch is based on 'origin/main', but the upstream is gone.
(use "git branch --unset-upstream" to fixup)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        bigfile.bin
        gitea-lfs-test/

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\31018\gitea-lfs-test> git push
Locking support detected on remote "origin". Consider enabling it with:
  $ git config lfs.http://localhost:3000/tangwenting/gitea-lfs-test.git/info/lfs.locksverify true
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 252 bytes | 252.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote: . Processing 1 references
remote: Processed 1 references in total
To http://localhost:3000/tangwenting/gitea-lfs-test.git
 * [new branch]      main -> main
PS C:\Users\31018\gitea-lfs-test> |

```

这里，我发现，git push 这条命令本身成功了，但是 bigfile.bin 还处于 **未跟踪状态**，没有被 add/commit 进去。原因是我目录里又多了一个 gitea-lfs-test/。

Here, I found that the git push command itself succeeded, but bigfile.bin is still in an untracked state and has not been added or committed. The reason is that an additional gitea-lfs-test/ directory appeared in my working directory.

```

PS C:\Users\31018\gitea-lfs-test> Remove-Item -Recurse -Force .\gitea-lfs-test
PS C:\Users\31018\gitea-lfs-test> dir

    目录: C:\Users\31018\gitea-lfs-test

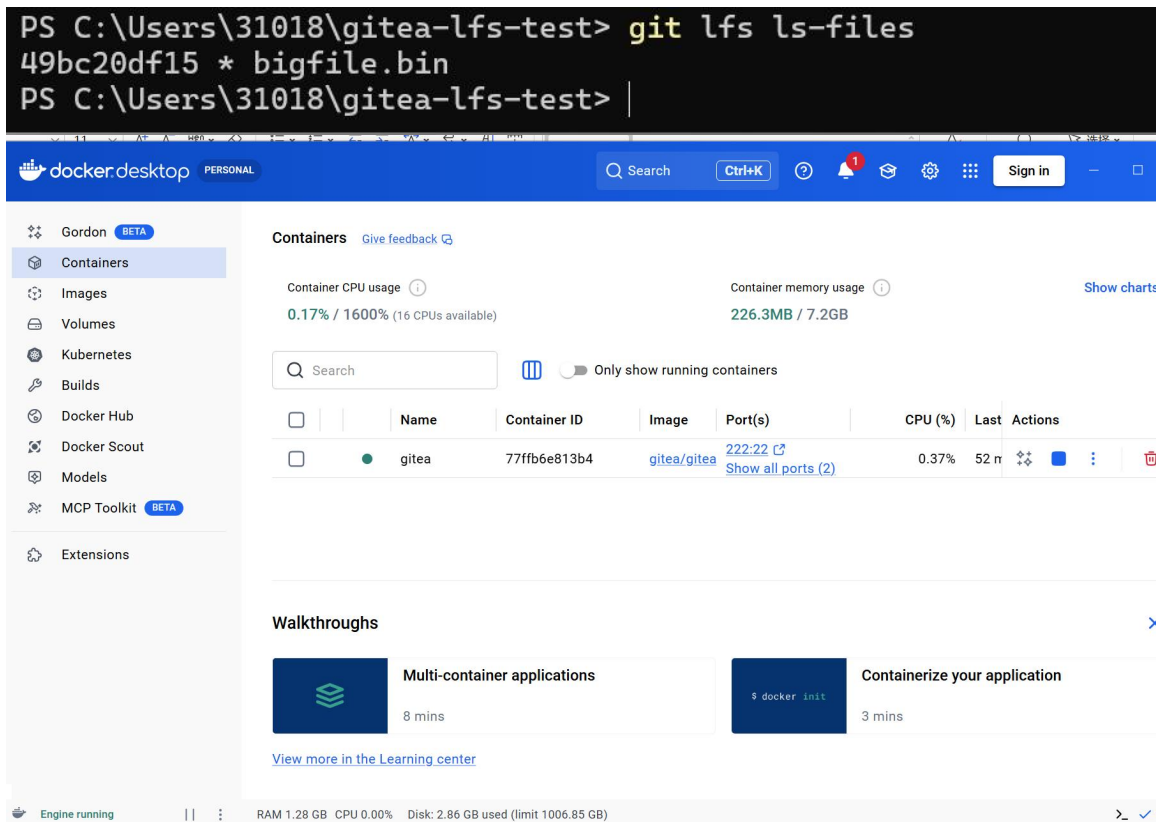
Mode                LastWriteTime         Length Name
----                -
-a----             2026/3/30      15:18           43 .gitattributes
-a----             2026/3/30      15:20    1073741824 bigfile.bin

PS C:\Users\31018\gitea-lfs-test> |

```

Then it is ok.

Step 9: Verify LFS (git lfs ls-files)



LFS is successful.

6. Conclusion and reflection

In this experiment, I successfully installed Docker Desktop and deployed Gitea locally using Docker. By accessing the local web interface, I was able to create and manage repositories. During the experiment, I learned how to use Git and Git LFS, and successfully created and uploaded a file larger than 1GB.

By executing commands such as `git lfs track` and `git lfs ls-files`, I verified that Gitea supports Git LFS. The results demonstrate that Gitea can effectively manage large files. This experiment enhanced my understanding of container technology, code hosting platforms, and large file version control mechanisms.